Reasoning test

For admission to Class 9

About the test

- 15 questions in 15 minutes: 5 each of maths, English and science, mixed together.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers

The questions test reasoning, not memory, using ideas taught up to Class 8. Chapters are from the NCERT (CBSE) books a student joining Class 9 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Ganita Prakash

- *Patterns in Mathematics* – number patterns and series
- *Perimeter and Area*
- *Prime Time* – primes, factors, HCF and LCM
- *Fractions*

Class 7 · Ganita Prakash

- *Working with Fractions* – fractions
- *Finding Common Ground* – primes, factors, HCF and LCM
- *Finding the Unknown* – simple equations

Class 7 · Curiosity (science)

- *Measurement of Time and Motion* – speed, distance and time

Class 8 · Ganita Prakash

- *A Square and A Cube* – squares and cubes
- *Proportional Reasoning-1* – ratio and proportion
- *Fractions in Disguise* – percentages, profit and loss, discounts, interest
- *Proportional Reasoning-2* – ratio and proportion
- *Tales by Dots and Lines* – averages (mean)

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Curiosity

- *Exploring Magnets* – magnets
- *A Journey through States of Water* – evaporation and condensation

Class 7 · Curiosity

- *Exploring Substances: Acidic, Basic and Neutral* – acids, bases and indicators
- *Electricity: Circuits and their Components* – electric circuits
- *Changes around Us: Physical and Chemical* – physical and chemical changes, rusting
- *Heat Transfer in Nature* – heat and conductors
- *Measurement of Time and Motion* – speed
- *Life Processes in Animals* – respiration
- *Life Processes in Plants* – photosynthesis and plant respiration
- *Light: Shadows and Reflections* – light, shadows and plane mirrors

Class 8 · Curiosity

- *Exploring Forces* – floating, sinking and density
- *Pressure, Winds, Storms, and Cyclones* – force and pressure, pressure in liquids
- *Particulate Nature of Matter* – evaporation and condensation
- *How Nature Works in Harmony* – food chains

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS

   How many seconds are there in 2.5 hours?

   A 150
   B 900
   C 9,000
   D 15,000

2. ENGLISH

   Choose the word most nearly opposite in meaning to "reluctant".

   A eager
   B slow
   C afraid
   D unwilling

3. SCIENCE

   In the food chain grass → grasshopper → frog → snake, which living thing is the producer?

   A Frog
   B Snake
   C Grasshopper
   D Grass

4. MATHS

   The ratio of boys to girls in a class is 4 : 5. If there are 20 boys, how many girls are there?

   A 16
   B 20
   C 24
   D 25

5. ENGLISH

   Choose the correct word: She has been living in Pune ___ 2019.

   A from
   B since
   C for
   D by

6. SCIENCE

   A card with a word printed on it is held in front of a plane mirror. In the mirror, the word appears

   A upside down
   B with left and right swapped
   C larger than on the card
   D smaller than on the card

7. MATHS

   When a number is doubled and then increased by 7, the result is 31. What is the number?

   Type-in answer: _____

8. ENGLISH

   When the coach said the match was "in the bag", he meant that

   A the team had packed its kit
   B the result was still uncertain
   C victory was almost certain
   D the match had been called off

9. SCIENCE

   Compared with the air we breathe in, the air we breathe out contains

   A more carbon dioxide
   B more oxygen
   C less water vapour
   D no carbon dioxide at all

10. MATHS

A mother and her daughter are 50 years old together. Five years ago, the mother was 7 times as old as the daughter. How old is the daughter now?

A 8
B 10
C 12
D 15

11. ENGLISH

Arrange the sentences in a logical order. P. Asha, their leader, checked the map. Q. The trekkers realised they were lost. R. They reached camp before dark. S. She found the right trail.

A PQSR
B QSPR
C QPRS
D QPSR

12. SCIENCE

A sharp knife cuts more easily than a blunt one pushed with the same force, because the sharp edge

A makes much more friction with what is cut
B pushes with a larger force overall
C acts on a smaller area, raising pressure
D spreads the force over a larger area

13. MATHS

A scooter worth ₹80,000 loses 10% of its value every year. What will it be worth after 2 years?

A ₹64,800
B ₹64,000
C ₹66,000
D ₹72,000

14. ENGLISH

Ali, Bina, Dev, Esha and Tara sit in a row of five. Tara is in the middle, Ali at the left end and Bina next to Ali. Dev is not next to Tara. Who is at the right end?

A Dev
B Esha
C Bina
D Cannot be determined

15. SCIENCE

A box weighing 600 N has one face of area 2 m^2 and another of 0.5 m^2. How many pascals more pressure does it exert resting on the small face than on the large face?

Type-in answer: _____

Answers

1	**C. 9,000**	One hour has 60 × 60 = 3,600 seconds, so 2.5 hours has 2.5 × 3,600 = 9,000 seconds. 150 is the number of minutes.
2	**A. eager**	Reluctant means unwilling or hesitant, so its opposite is eager. "Unwilling" is a word with the same meaning, not the opposite.
3	**D. Grass**	A producer makes its own food by photosynthesis, and in a food chain it always comes first. Grass is the producer; the others are consumers that eat.
4	**D. 25**	20 boys is 4 parts, so one part is 5 students. Girls are 5 parts, which is 25. 16 comes from using the ratio the wrong way round.
5	**B. since**	"Since" is used with a point in time (since 2019); "for" is used with a length of time (for five years).
6	**B. with left and right swapped**	A plane mirror forms an image the same size and the same way up, but with left and right swapped, which is why the word reads backwards.
7	**12**	Work backwards: 31 − 7 = 24, and 24 ÷ 2 = 12. Check: 12 × 2 + 7 = 31. This is a type-in question.
8	**C. victory was almost certain**	"In the bag" is an idiom meaning that success is as good as secured. It has nothing to do with an actual bag.
9	**A. more carbon dioxide**	The body uses oxygen to release energy from food and gives off carbon dioxide, so breathed-out air holds more carbon dioxide and less oxygen.
10	**B. 10**	Five years ago their ages added to 40, in the ratio 7 : 1, so they were 35 and 5. The daughter is now 10 and the mother 40.
11	**D. QPSR**	The problem comes first (Q), then the response (P), then the solution (S), then the result (R). "She" in S needs Asha named before it.
12	**C. acts on a smaller area, raising pressure**	Pressure is force divided by area. The same force on the thin edge of a sharp knife gives a much greater pressure, which cuts.
13	**A. ₹64,800**	After one year it is worth ₹72,000. The second year takes 10% of ₹72,000, which is ₹7,200, leaving ₹64,800. ₹64,000 takes ₹8,000 off twice.
14	**A. Dev**	Seats 1, 2 and 3 are Ali, Bina and Tara. Dev and Esha take seats 4 and 5; seat 4 is next to Tara, so Dev must be in seat 5.
15	**900**	On the small face: 600 ÷ 0.5 = 1,200 Pa. On the large face: 600 ÷ 2 = 300 Pa. The difference is 900 Pa. This is a type-in question.